SEED: Agents Learn Rules From Their Own Completed Attempts

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Researchers from five universities released SEED, short for Self-Evolving On-Policy Distillation, a training method that turns an AI agent's completed attempts into guidance for its next update. Standard reinforcement learning often gives one score after a long run, so the model learns little about which individual choices helped. SEED asks the current model to summarize each finished run as a short rule, then distils that rule back into the policy by measuring how

it shifts the probability of the same action tokens. After each update, the improved policy becomes both the next actor and the next skill analyser. This supplies feedback at each decision without keeping the rules at deployment. On ALFWorld, a household-task simulator, SEED raised the Qwen2.5-3B average from 75.0% with Group Relative Policy Optimization (GRPO) to **91.8%**. On WebShop, success rose from **63.3% to 78.9%**.

An agent that fails after 40 actions needs more than a final zero. SEED can point training toward an earlier bad search or missed observation, while the deployed model still uses an ordinary prompt. The authors also report that SEED reached **80.7% with 60% of the ALFWorld training data**, beating GRPO's 75.0% with the full set. These are paper results, not an independent reproduction.

The setup is not cheap or broadly proven. Its first stage uses 1,440 runs labelled through an external GLM-5.2 endpoint, and the published recipe assumes eight GPUs. The only released checkpoint is specialised for ALFWorld and uses the Qwen Research Licence, although the training code is MIT-licensed. Unlike LatentSkill, SEED absorbs the rules into the main model rather than storing them in swappable adapters.

Sources:

- SEED paper
- SEED project page
- SEED code and training recipe
- Released ALFWorld model card

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